PRAKTIKUM SISTEM OPERASI

MODUL 9

SYSCALL XINU

Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

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JURNAL/TP

1. Buat syscall baru seperti yang ditunjukkan pada modul syscall poin 9.5! (sertakan Screenshot kode dan hasil run)
   a. Membuat syscal baru

```c
// Nama: Aulia Jasifa Br Ginting
// NIM: 2311104060

#include <xinu.h>

/*------------------------------------------------------------------------
 *  chname  -  Dummy syscall
 *------------------------------------------------------------------------
 */
syscall chname(pid32 pid, char *newname) {
        intmask mask;

        mask = disable();

        /* Validasi PID*/
        if (isbadpid(pid)) {
                restore(mask);
                return SYSERR;
        }

        /* Validasi nama*/
        if (newname == NULL) {
                restore(mask);
                return SYSERR;
        }

        /* Ubah nama proses*/
        strncpy(proctab[pid].prname, newname, PNMLEN-1);
        proctab[pid].prname[PNMLEN-1] = '\0';

        restore(mask);
        return OK;
}
```

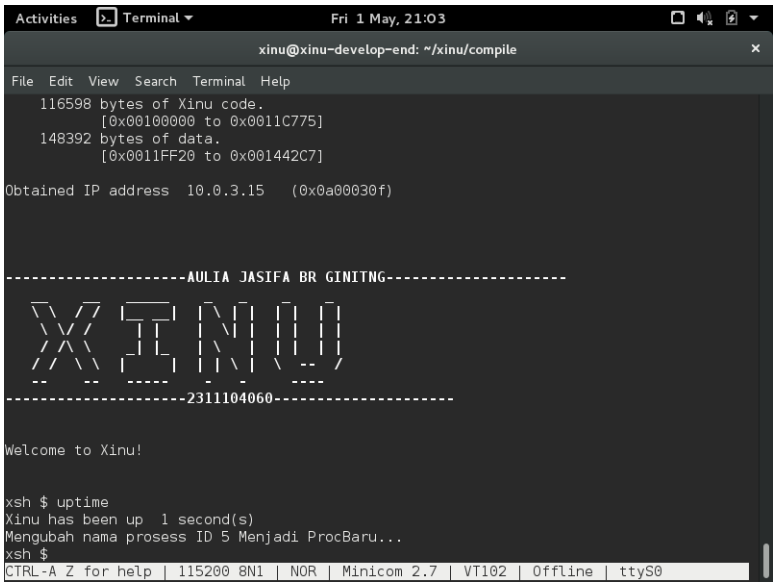
   b. Menambahkan prototypes.h

```c
extern  status  ascdate(uint32, char *);

/* in file bufinit.c */
extern  status  bufinit(void);

/* in file chprio.c */
extern  syscall chprio(pid32, pri16);

/* in file chname.c */
extern  syscall chname(pid32, char *);

/* in file clkupdate.S */
extern  uint32  clkcount(void);

/* in file clkhandler.c */
extern  interrupt clkhandler(void);

/* in file clkinit.c */
extern  void    clkinit(void);

/* in file clkdisp.S */
extern  void    clkdisp(void);

/* in file close.c */
extern  syscall close(did32);

/* in file control.c */
extern  syscall control(did32, int32, int32, int32);

/* in file create.c */
extern  pid32   create(void *, uint32, pri16, char *, uint32, ...);

/* in file ctxsw.S */
extern  void    ctxsw(void *, void *);
```

c. xsh_uptime

```c
/* xsh_uptime.c - xsh_uptime */

#include <xinu.h>
#include <stdio.h>
#include <string.h>

/*------------------------------------------------------------------------
 * xsh_uptime - shell to print the time the system has been up
 *------------------------------------------------------------------------
 */
// Aulia Jasifa Br Ginting
// 2311104060
shellcmd xsh_uptime(int nargs, char *args[])
{
        uint32  days, hrs, mins, secs;  /* Days, hours, minutes,  and   */
                                        /*  seconds since system boot   */
        uint32  secperday = 86400;      /* Seconds in a day             */
        uint32  secperhr  =  3600;      /* Seconds in an hour           */
        uint32  secpermin =    60;      /* Seconds in a minute          */

        /* Nama proses baru */
        char prosesBaru[] = "ProcBaru";

        /* For argument '--help', emit help about the 'uptime' command  */
        if (nargs == 2 && strncmp(args[1], "--help", 7) == 0) {
                printf("Use: %s\n\n", args[0]);
                printf("Description:\n");
                printf("\tDisplays time since the system booted\n");
                printf("Options:\n");
                printf("\t--help\t display this help and exit\n");
                return 0;
        }

        /* Subtract number of hours */
        hrs   = secs/secperhr;
        secs -= hrs*secperhr;

        /* Subtract number of minutes */
        mins  = secs/secpermin;
        secs -= mins*secpermin;

        printf("Xinu has been up ");
        if (days > 0) {
                printf(" %d day(s) ", days);
        }

        if (hrs > 0) {
                printf(" %d hour(s) ", hrs);
        }

        if (mins > 0) {
                printf(" %d minute(s) ", mins);
        }

        if (secs > 0) {
                printf(" %d second(s) ", secs);
        }
        printf("\n");

        /* Syscall baru*/
        kprintf("Mengubah nama prosess ID 5 Menjadi %s...\n", prosesBaru);
        chname(5, prosesBaru);

        return 0;
}
```

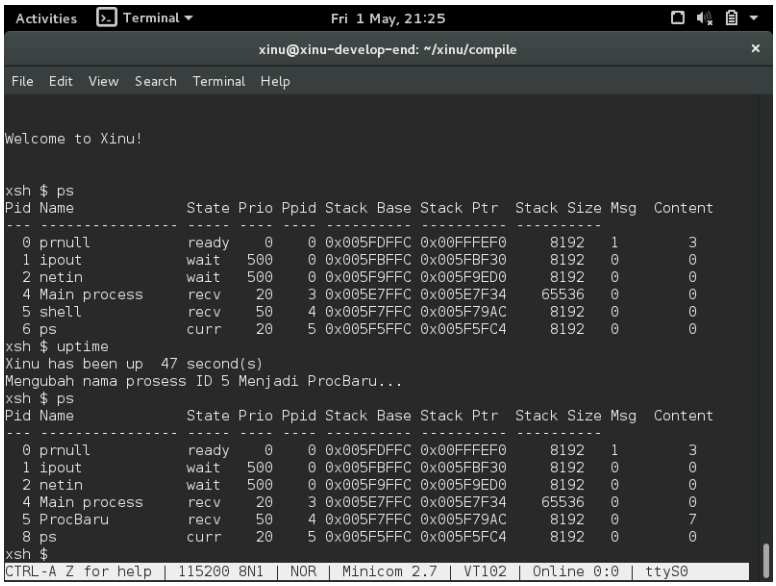
d. Output



2. Perbaiki syscall chprio (xinu/system/chprio.c) dengan memperhatikan validasi input
   - Pastikan id adalah angka dari 0 – NPROC (ukuran maks banyaknya proses)
   - Pastikan prioritas adalah bilangan yang positif

   Compile dan jalankan Xinu dengan syscall yang telah diperbaiki
   - make
   - clean

```c
// Nama: Aulia Jasifa Br Ginting
// NIM: 2311104060

#include <xinu.h>

/*------------------------------------------------------------------------
 *  chprio  -  Change the scheduling priority of a process
 *------------------------------------------------------------------------
 */
syscall chprio(pid32 pid, pri16 newprio)
{
        intmask mask;
        struct procent *prptr;
        pri16 oldprio;

        mask = disable();

        if (pid < 0 || pid >= NPROC) {
                restore(mask) ;
                return SYSERR;
        }

        if (newprio <= 0) {
                restore(mask);
                return SYSERR;
        }

        prptr = &proctab[pid];
        oldprio = prptr->prprio;
        prptr->prprio = newprio;

        restore(mask);
        return oldprio;
```



3. Lakukan hal-hal berikut ini

   Edit xsh_uptime.c

   - Tambahkan kode berikut

```
        /* Subtract number of hours */
        hrs   = secs/secperhr;
        secs -= hrs*secperhr;

        /* Subtract number of minutes */
        mins  = secs/secpermin;
        secs -= mins*secpermin;

        printf("Xinu has been up ");
        if (days > 0) {
                printf(" %d day(s) ", days);
        }

        if (hrs > 0) {
                printf(" %d hour(s) ", hrs);
        }

        if (mins > 0) {
                printf(" %d minute(s) ", mins);
        }

        if (secs > 0) {
                printf(" %d second(s) ", secs);
        }
        printf("\n");

        /* Syscall baru*/
        kprintf("Mengubah prioritas proses ID 5 menjadi 33...\n");
        chprio(5, 33);

        return 0;
}
```

   Compile source code tersebut dengan perintah

   - make clean
   - make

   Jalankan perintah ps

   - xsh $ ps

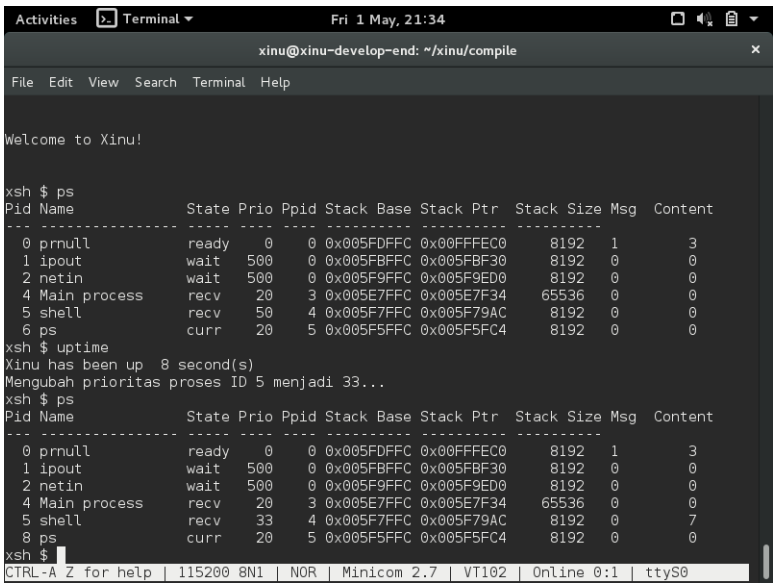
- perhatikan prioritas proses dengan id = 5

Jalankan uptime

- xsh $ uptime
- Perhatikan hasil perintah tersebut

Jalankan ps

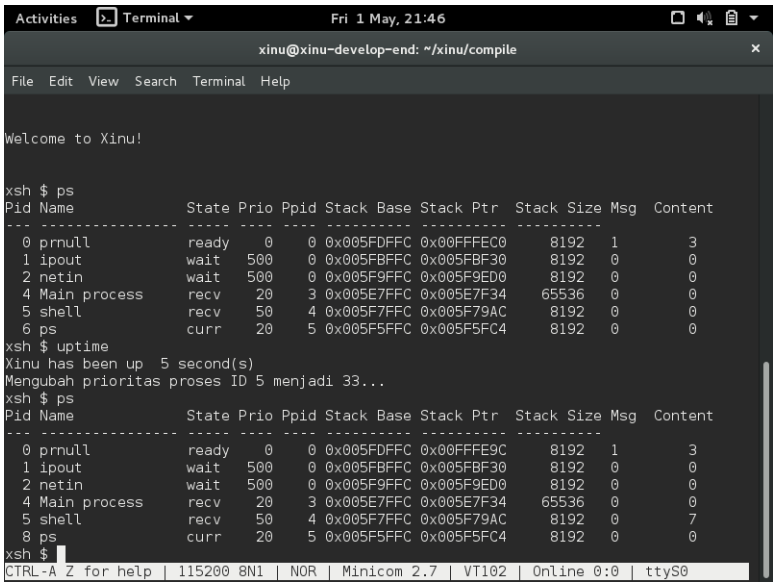
- xsh $ ps
- perhatikan prioritas proses dengan id = 5 seharusnya sudah berubah



Testing chprio syscall yang telah diubah

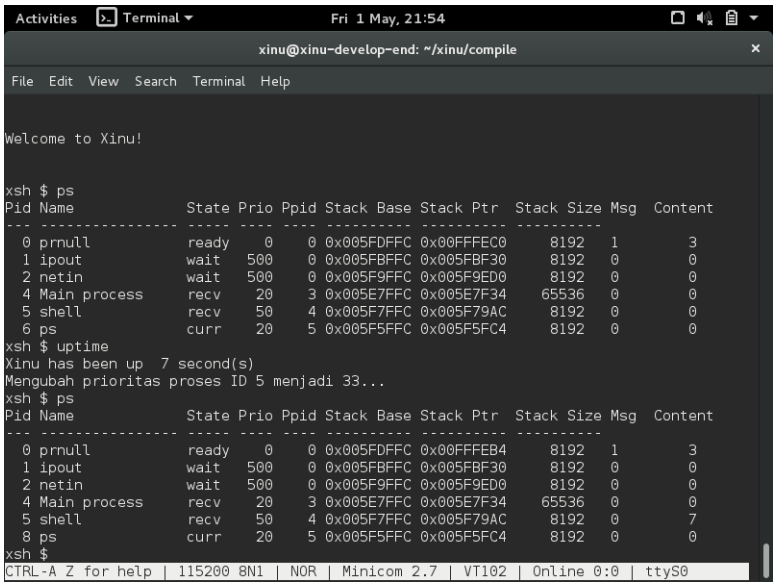
- Testing prioritas tidak boleh < 0: Ubah “chprio(5,33)” menjadi “chprio(5,-3)” pada xsh_uptime.c

```
        /* Subtract number of hours */
        hrs   = secs/secperhr;
        secs -= hrs*secperhr;

        /* Subtract number of minutes */
        mins  = secs/secpermin;
        secs -= mins*secpermin;

        printf("Xinu has been up ");
        if (days > 0) {
                printf(" %d day(s) ", days);
        }

        if (hrs > 0) {
                printf(" %d hour(s) ", hrs);
        }

        if (mins > 0) {
                printf(" %d minute(s) ", mins);
        }

        if (secs > 0) {
                printf(" %d second(s) ", secs);
        }
        printf("\n");

        /* Syscall baru*/
        kprintf("Mengubah prioritas proses ID 5 menjadi 33...\n");
        chprio(5, -3);

        return 0;
}
```



- Testing id adalah valid: Ubah “chprio(5,33)” menjadi “chprio(3000,3)”

```
        /* Subtract number of hours */
        hrs   = secs/secperhr;
        secs -= hrs*secperhr;

        /* Subtract number of minutes */
        mins  = secs/secpermin;
        secs -= mins*secpermin;

        printf("Xinu has been up ");
        if (days > 0) {
                printf(" %d day(s) ", days);
        }

        if (hrs > 0) {
                printf(" %d hour(s) ", hrs);
        }

        if (mins > 0) {
                printf(" %d minute(s) ", mins);
        }

        if (secs > 0) {
                printf(" %d second(s) ", secs);
        }
        printf("\n");

        /* Syscall baru*/
        kprintf("Mengubah prioritas proses ID 5 menjadi 33...\n");
        chprio(3000, 3);

        return 0;
}
```



- Hasil dua testing di atas adalah prioritas tidak berubah karena salah argument (dibuktikan dengan menggunakan perintah ps)